UNITED INTERNATIONAL UNIVERSITY

Department of Computer Science and Engineering (CSE)

Course Syllabus

1	**Course Title**	Fundamentals of Calculus
2	**Course Code**	Math-1151/ Math-151
3	**Trimester and Year**	Spring-2024
4	**Pre-requisites**	
5	**Credit Hours**	3
6	**Section**	
7	**Class Hours**	**Section AA:** Saturday & Tuesday: 11:10 AM-12:30 PM **Section R:** Sunday & Wednesday: 12:30 PM-1:50 PM
8	**Class Room**	
9	**Instructor's Name**	S. M. Yiasir Arafat (SMYA)
10	**Email**	yiasirarafat@gmail.com; Mobile - 01521471924
11	**Office**	Room # 0310
12	**Counselling Hours**	Saturday & Tuesday: 9:50 AM – 11:10 AM Sunday & Wednesday: 9:50 AM – 11:10 AM
13	**Text Book**	***Calculus 10-th Edition - by Howard Anton, Irl Bivens and Stephen Davis***
14	**Reference**	
15	**Course Contents (approved by UGC)**	Function, Domain and Range of a Function, Translation, reflection, compression and stretches of a function. Even and Odd functions, Inverse functions, One to One and many to one function, Family of Exponential, logarithmic, sine and cosine function, Limit, continuity and differentiability, Tangent line, Derivative and Chain rule, An overview of area problem, Newton's anti-derivative method in finding area, Indefinite integral, fundamental theorem of calculus, Definite integral, Area between two curves, arc length.
16	**Course Outcomes (COs)**	see table below

COs	**Description**
CO1	Define relation of two variables, interval, function, domain and range of different types of function, Composite, odd and even functions, inverse functions, family of functions, Limit, continuity and differentiability of various functions.
CO2	Analyze graph of functions (Translations, reflections, Stretch and compress), limits, continuity and differentiability.
CO3	Compute differentiations and integrations of functions.
CO4	Apply differentiation to find the rate of change of functions, velocity of a moving particle, average velocity, instantaneous velocity and slope of secant and tangent line.
CO5	Interpret the relationship of definite integral with area problem, indefinite integral and arc length.

17	**Teaching Methods**	Lecture, Case Studies, Assignments.
18	**CO with Assessment Methods**	

CO	Assessment Method	(%)
-	Attendance	5
1-5	Assignments	5
1-5	Class Tests	20
×	Presentation	10
1-2	Midterm exam	30
3-5	Final exam	40

19 Mapping of COs and Program outcomes

COs	Program Outcomes(POs)											
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	√											
CO2	√											
CO3	√											
CO4	√											
CO5		√										

20 Lecture Outline

Class	Topics/Assignments	COs	Reading Reference	Lecture Outcomes/Activities
1-2	Function, Domain and Range of a Function	1	Chap: **0.1** Exercise: 1 , 3, 7, 9	Q/A Test, Assignment
3-7	Translation, reflection, Even and Odd function, Inverse function, One to One, many to one function and Family of Exponential & logarithmic function	1,2	Chap: **0.2** Exercise: 5 – 24 Chap: **0.4** Exercise: 1, 3 Chap: **0.5** Page 53-Figure 0.5.2, Page 54-example-1, Page 56-Figure-0.5.6.	Q/A Test, Assignment
8-12	Limit, continuity and differentiability and Tangent line, Tangent line	1,2 ,4	Chap: **1.1** Example-5, 7; Exercise: 1 – 8 Chap: **1.5** Example-2, 3, 4; Exercise-11,13,15,17,19 Chap: **2.2** Exercise-47, 48. Chap: **2.1** Exercise:11 – 18	Q/A Test
MID Examination				

13-14	Differentiation of different types of functions (Derivative and Chain rule)	3	Chap: **2.6** Exercise: 7 – 49 (odd)	Q/A Test, Assignment
15	An overview of area problem, Newton's anti-derivative method in finding area,	4	Chap: **5.1** Example: 1, 2; Exercise: 13 – 18	Q/A Test, Assignment
16-20	Indefinite integral, fundamental theorem of calculus, Definite integral, Area between two curves,	4,5	Chap: **5.5** Exercise: 13 – 20 Chap: **5.2** Exercise: 5 – 8, 15 – 30. Chap: **5.3** Exercise: 15 – 56 Chap: **5.6** Exercise:13 – 30, Chap: **6.1** Exercise:1 – 7, 11 – 12	Q/A Test, Assignment
21-23	Different types of Integration (Principles of Integral evaluation, Integration by parts, Trigonometric Substitution).	5	**7.1:** A review of familiar integration formulas: 1-13, 21-26 Exercise Set 7.1: 1-30 **7.2:** Examples: 1-8 Exercise Set 7.2: 1-38 **7.4:** Examples: 1-6 Exercise Set 7.4: 1-26, 37-47 (Only odd numbers)	Q/A Test
24	**Review Class**			
FINAL Examination				

***** Some Important Dates: 7th class (CT-1), 11th class (CT-2), 19th class (CT-3), 23rd class (CT-4)**

Appendix 1: Assessment Methods

Assessment Types	Marks
Attendance	5%
Assignments	5%
Class Tests	20%
Mid Term	30%
Final Exam	40%

Appendix 2: Grading Policy

Letter Grade	Marks %	Grade Point	Letter Grade	Marks%	Grade Point
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Appendix-3: Program outcomes

POs	Program Outcomes
PO1	An ability to apply knowledge of mathematics, science, and engineering
PO2	An ability to identify, formulate, and solve engineering problems
PO3	An ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability
PO4	An ability to design and conduct experiments, as well as to analyze and interpret data
PO5	An ability to use the techniques, skills, and modern engineering tools necessary for engineering practice
PO6	The broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context
PO7	A knowledge of contemporary issues
PO8	An understanding of professional and ethical responsibility
PO9	An ability to function on multidisciplinary teams
PO10	An ability to communicate effectively
PO11	Project Management and Finance
PO12	A recognition of the need for, and an ability to engage in life-long learning